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Spent mushroom compost as substrate for the production of industrially important hydrolytic enzymes by fungi *Trichoderma* spp. and *Aspergillus niger* in solid state fermentation

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ABSTRACT

Mushroom production is the biggest solid state fermentation industry in the world. Disposal and storage of spent mushroom compost (SMC) that remains after mushroom harvest poses a big economic and environmental problem. Production of industrially important hydrolytic enzymes by fungi on various agro-industrial wastes is a significant, open chapter in biotechnology. This paper proposes a novel application of SMC as substrate for cultivation of fungi in solid state fermentation (SSF) in order to obtain the enzymes cellulase, xylanase, amylase and β-glucosidase. SMC can be used as a good substrate for cultivation of *Trichoderma* and *Aspergillus* without the addition of supplementary (nutritive) elements. Starting amount of SMC was reduced by 30% due to hydrolysis by a complex of cellulolytic enzymes. Material that is left behind is a more suitable fertilizer for horticulture. One fungal isolate was pointed out as a promising producer (*Trichoderma atroviride* isolate T42). It produced the greatest amount of total protein (0.204 mg mL^{-1}), five isoforms of β-glucosidase and the highest level (12 isoforms) of both endocellulase (0.76 U mL^{-1}) and xylanase (2.31 U mL^{-1}). The capacity of T42 to produce all examined enzymes in such a high number of isoforms demonstrates successful adaptation to new substrates.

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1. Introduction

Mushroom production is the biggest solid state fermentation (SSF) industry in the world (Moore and Chiu, 2001). As much as 7,959,979 tonnes of mushrooms were produced in 2012, with China accounting for most of the production (5,150,000 tonnes), while 1,869,091 tonnes were harvested in Europe (FAO, 2013). The white button mushroom, *Agaricus bisporus* (Lange) Imbach, is the most commonly cultivated and consumed mushroom species in the world and in Serbia. Substrate for mushroom growing is pasteurized waste, made from straw (60–70%), poultry or horse manure (28–34%), gypsum (2–4.5%), hay and corn stover.

Disposal and storage of spent mushroom compost (SMC) obtained as residue after mushroom harvest has become a major environmental problem today. Approximately 5 kg of SMC is produced for every 1 kg of grown mushrooms (Finney et al., 2009). Besides the fact that disposal and storage of SMC is an economic problem, this also causes a considerable environmental problem due to excess nitrate and phosphorous. Current disposal strategies for SMC include burning, spreading on land, composting with animal manure or land filling. Burning is not environment-friendly, and land filling is not suitable for extensive use in food production and it must be controlled because of N and P levels (Phan and Sabaratnam, 2012). There are some trials to reuse SMC as fertilizer for gardens. However, this requires durable pre-treatment and processing in small charges. One of the main difficulties in the management of SMC is the lack of waste management supervision in many countries. An evident way to overcome this problem and find a solution is to explore and develop new applications of SMC. In this regard, the number of publications related to the subject has somewhat increased in recent years. One of the investigated and possible solutions is enzyme recovery from SMC, as well as bioremediation, animal feed, energy feedstock and crop production (Jordan and Mullen, 2007; Phan and Sabaratnam, 2012). Some of

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these possibilities are not feasible (e.g. animal feed) (Adamović et al., 1998), others are demanding or difficult to implement (Finney et al., 2009), and the rest are still subject of research at present. None of them have been applied in the industry yet. Usage of SMC as a substrate for fungal enzyme production has not been reported in scientific literature or in the industry to this day.

SMC is still rich in nutrition components after mushroom cultivation. It contains cellulose 38%, hemicellulose 19%, lignin 25% and other organic matter 645 g kg^{-1} of SMC (Jordan et al., 2008). Based on high content of cellulose and hemicellulose (approximately 57%) SMC can be used as feedstock for cultivation of fungi producing cellulolytic enzymes. Micromycetes *Trichoderma* and *Aspergillus* are the most important producers of cellulase, amylase and hemicellulase enzymes for industrial purposes (Pandey et al., 1999; Kubicek, 2013). The aim of this paper is to open a new chapter of solutions to overcome problems with disposal of SMC by using it as a substrate for cultivation of fungi in SSF and, thus, for recovery of these enzymes. SSF is widely accepted for fungal enzyme production (Pandey et al., 2008; Singhania et al., 2010). These enzymes are vital due to their involvement in bioethanol production, amylase in the first generation of bioethanol production and cellulase complex and xylanase in the second (Cardona and Sanchez, 2007). Furthermore, they are also used in paper, food, textile, and detergents industries. Production of enzymes requires/includes testing of various types of substrate and the usage of waste materials (as substrate) is dominant.

The aim of this work was to examine the usage of SMC, without pre-treatment or supplementation with any nutritional elements, as a substrate for cultivation of *Trichoderma* spp. grown for the production of cellulases, xylanases and amylases. Examined isolates of *Trichoderma* spp. were compared with *Aspergillus niger* ATCC 10,864, a positive control as a reference producer of these enzymes.

2. Material and methods

2.1. Chemical abstract

All reagents and solvents used in this study were of the highest available purity and were purchased from Merck (Darmstadt, Germany), Sigma-Aldrich (St. Louis, MO, USA), Fermentas (Vilnius, Lithuania) and Metabion international AG (Germany).

2.2. Microorganisms

Six fungal isolates were used, as described in Table 1. Three isolates of *Trichoderma* spp. with laboratory base designation UB483TK1, UB483TH2, UB483TA1 (marked in the text below as TK1, TH2 and TA1, respectively), originated from wild types of corn (*Zea mays* L.) kindly donated by the Maize Research Institute Zemun Polje, Belgrade. The other three *Trichoderma* isolates, T42, T10 and T55 were obtained from white button mushroom (*Agaricus bisporus* (Lange) Imbach) or shiitake (*Lentinus edodes* (Berk) Singer) fruiting bodies, of which T10 was already identified and characterized as *Trichoderma harzianum* (Kosanović et al., 2013). Hitherto uncharacterized strains (TK1, TH2, TA1, T42 and T55) were morphologically and molecularly characterized in this work. *A. niger* ATCC 10,864 (laboratory base designation UB483AN43, marked in the text below as AN43) was chosen as recognized and standard producer of the examined enzymes and purchased from Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ).

All strains were cultivated on potato dextrose agar (PDA) at 28 °C for 7 days to obtain matured spores. Spore suspensions were prepared in 0.1% Tween 80 solution at a concentration of 6.0×10^5spores mL^{-1}.

2.3. Morpho-physiological studies

Morpho-physiological observations of the isolates were made from day 2 to day 6 of cultivation on PDA at 20 °C or 25 °C. Isolate identification was based on colony colour, pigmentation of reverse side and odour; sporulation type, conidia shape, colour, and dimension; growth rate and optimal growth temperature. Growth rate of the isolates was defined by measuring colony diameter every day for 72 h. The effect of temperature on growth was studied on isolates grown on PDA in darkness at 15, 20, 25, 30 and 35 °C for 4 days.

2.4. Molecular identification

All tested isolates were grown on PDA at 25 °C. DNA extraction was performed following the CTAB protocol (Day and Shattock, 1997). Polymerase chain reaction (PCR) assays were carried out with the universal primer pair ITS1/ITS4 (White et al., 1990). PCR reaction mix contained 20 ng DNA template, 1 × PCR Master Mix (Fermentas, Vilnius, Lithuania) and 0.4 μM of each primer. Thirty five PCR cycles were performed for amplification (White et al., 1990). PCR products were separated in 1% agarose gel and amplified products were purified using mi-PCR Purification Kit (Metabion international AG, Germany) and sequenced in both directions using forward ITS1 and reverse ITS4 primers. Sequences were assembled using Pregap4 from the Staden program package (Staden et al., 2000) and deposited in the NCBI GenBank under accession numbers given in Fig. 1. The obtained sequences were aligned with sequences of *Trichoderma* spp. isolates, already identified as reference (Kosanović et al., 2013) by Clustal X (Thompson et al., 1997) in MEGA version 6 (Tamura et al., 2013). A phylogenetic tree was constructed by MEGA version 6, and analysed by the Maximum Parsimony method based on a close-neighbour-interchange algorithm (Nei and Kumar, 2000). Initial phylogenetic trees were obtained by searching level 5, through random addition of sequences (10 replicates). One of the eight most frugal trees is shown in Fig. 1.

2.5. Solid state fermentation (SSF)

Raw, untreated, SMC was obtained after harvesting of *A. bisporus*, from a local mushroom farm and used as substrate for

Table 1
Strains of *Trichoderma* spp. used for enzyme production on spent mushroom compost.

Isolate code	Host	Origin of isolate	Year of collection
TK1	Corn (*Zea mays*)	Zrenjanin	2010
TH2	Corn (*Zea mays*)	Paraćin	2010
TA1	Corn (*Zea mays*)	Kruševac	2010
T42	White button mushroom (*Agaricus bisoporus*)	Veliko Gradište	2007
T10	White button mushroom (*Agaricus bisoporus*)	Požarevac	2006
T55	Shiitake (*Lentinus edodes*)	Ugrinovci	2008

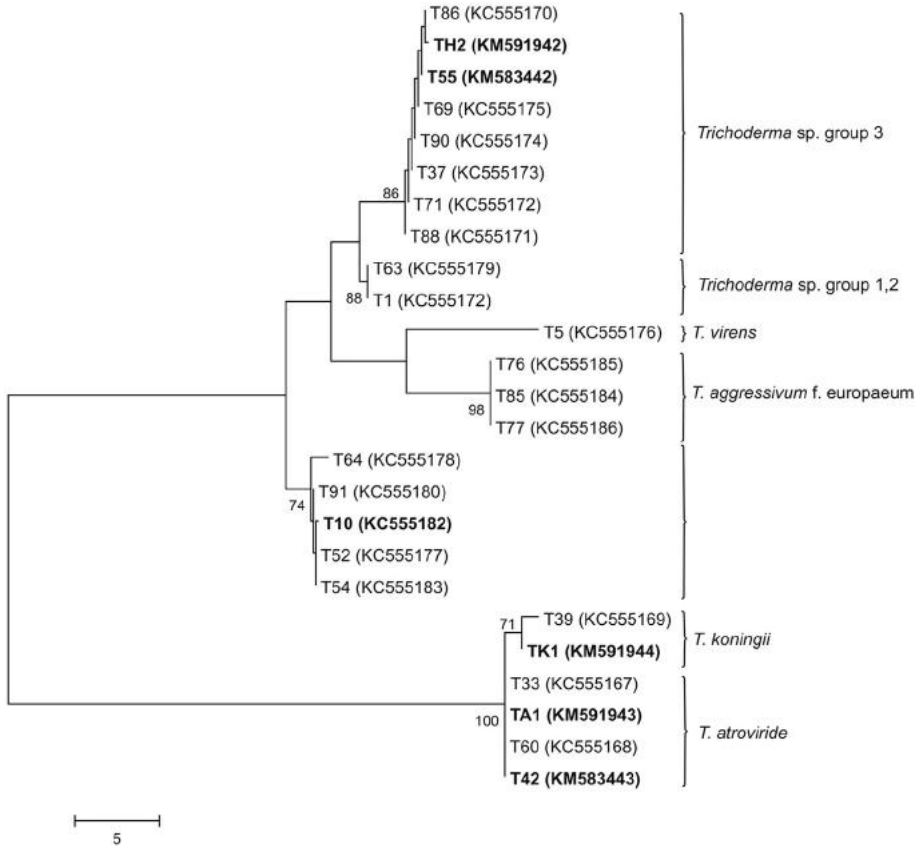

Fig. 1. Phylogenetic tree constructed by parsimony analyses of ITS1/ITS4 sequences of 25 *Trichoderma* isolates. Numbers on the branches are bootstrap values obtained for 1000 replicates (only values above 70% are shown). Previously identified strains are described by Kosanović et al. (2013), strains analyzed in this paper are shown in bold. The tree is drawn to scale; with branch lengths calculated using the average pathway method and are in the units of the number of changes over the whole sequence.

SSF without prior treatment. SSF was carried out in a 250 mL Erlenmeyer flask covered with microbiological cotton wool caps. Each Erlenmeyer flask contained 50 g of substrate moistened with distilled water to achieve relative humidity of 60% (calculated), autoclaved and aseptically inoculated. Spore suspension was spread on substrate in a 1:15 ratio (w/V). SSF was carried out in duplicates at 28 °C for 7 days.

2.6. Enzyme extraction

Fermentation was terminated by adding cold buffer solution (50 mM acetate buffer pH 5.0) in a 1:2 ratio w/V and homogenized with IkaTurrax homogenizer in 3 cycles of 15 s. Extraction was carried out at room temperature for 1 h. The obtained mixture was centrifuged at 5000 *g* for 15 min. Supernatant, in the form of crude fermentation extract (FE) was used for further analysis.

2.7. Analysis of SMC during fungal SSF

2.7.1. Microscopic analysis of wheat straw

Light microscopy (LeitzDiaplan microscope) with a 250 times magnification was used for monitoring of changes on wheat straw surface during the cultivation of fungi. Raw wheat straw, wheat straw in SMC after growing of *A. bisporus*, and wheat straw after growing of *Trichoderma* spp. and *A. niger* used in this study were analysed by microscopy. A TCA-1.31C microscopic camera was used for recording.

2.7.2. Parameters of fermentation efficiency

Concentrations of reducing sugars obtained by SSF were determined using dinitrosalicylic acid (DNS) reagent (Bernfeld, 1955). The reaction was stopped by addition of DNS reagent, followed by boiling for 5 min. After cooling, the absorbance of the reaction mixture was measured at 540 nm. Glucose was used as standard.

Protein concentrations were determined by Coomassie Brilliant Blue G-250 (Bradford, 1976) using bovine serum albumin as protein standard.

2.8. Production of enzymes and zymographic detection of isoforms

2.8.1. Endocellulase production

Carboxymethyl cellulose (CMC) was used as substrate to determine endocellulase activity. Samples were incubated with 2.0% (w/V) CMC in 50 mM acetate buffer pH 5.0 at 50 °C for 60 min. Reducing sugars released during the reaction were determined as described in Section 2.7.2. Glucose was used as standard. One unit (U mL^{-1}) of endocellulase activity was defined as the amount of enzyme required to produce 1 μmol of glucose in 1 min at 50 °C. Specific endocellulase activity (U mg^{-1}) was expressed as 1 U per 1 mg of protein present in the FE.

Proteins in the FE were separated by isoelectric focusing (IEF) using the Multiphor II electrophoresis system (Pharmacia-LKB Biotechnology) according to the manufacturer's instructions. Focusing was carried out on a 6% polyacrylamide gel with ampholytes in the 3–10 pH range. Endocellulase isoforms were detected in acrylamide (AA) (7.5%) gel with copolymerized 1% CMC, previously printed from IEF gel for 55 min. The cellulase zymogram

was further prepared as previously described by Pavlović et al. (2012).

2.8.2. β-d-glucosidase production

β-d-glucosidase activity was determined using *p*-nitrophenyl-β-d-glucopyranoside (*p*NPG) ($\varepsilon = 18{,}5\ cm^2\mu mol^{-1}$) as substrate. Test samples were incubated with 0.55 mM pNPG in 50 mM acetate buffer pH 5.75 at 37 °C for 30 min. The reaction was stopped by 1M sodium carbonate. The absorbance of released *p*-nitrophenol was measured at 405 nm. One unit ($U\ mL^{-1}$) of β-D glucosidase activity was defined as the amount of enzyme required to produce 1 μmol of *p*-nitrophenol per 1 min at 37 °C. Specific β-glucosidase activity ($U\ mg^{-1}$) was expressed as 1 U per 1 mg of protein present in the FE.

β-glucosidase isoforms were detected on nitrocellulose membrane (NC) after printing from IEF gel for 20 min. IEF was performed as described in Section 2.8.1. β-glucosidase isoforms were visualized as previously described by Kwon et al. (1994).

2.8.3. Xylanase production

Xylanase activity was determined using 1% xylan solution in 50 mM acetate buffer pH 5.0, as substrate. The reaction was carried out for 15 min at 50 °C. Reducing sugars released during the reaction were determined as described in Section 2.7.2. Xylose was used as standard. One unit of xylanase activity was defined as the amount of enzyme required to produce 1 μmol of xylose per 1 min at 50 °C. Specific xylanase activity ($U\ mg^{-1}$) was presented as 1 U per 1 mg of proteinpresent in the FE.

Xylanase isoforms were detected after incubation of the IEF gel in 1% xylan solution in 50 mM acetate buffer pH 5.0 for 1 h at 37 °C. IEF was performed as described in section 2.8.1. The gel was coloured with 0.1% Congo red, and then washed in 1 M NaCl. Xylanase isoforms were detected on red gel as illuminated bands. Prior to scanning, the gel was rinsed in acetic acid for 3 min resulting in change of background colour from red to blue-violet.

2.8.4. Amylase production

Amylase activity was assayed at pH 5.0 according to the DNS procedure (Bernfeld, 1955) the use of 1.0% (w/V) soluble starch as substrate for 15 min at 55 °C. Maltose was used as standard. One unit of α-amylase activity ($U\ mL^{-1}$) was defined as the amount of enzyme required to produce 1 μmol of maltose in 1 min at 55 °C. Specific amylase activity ($U\ mg^{-1}$) was expressed as 1 U per 1 mg of protein present in the FE.

The amylase zymogram was prepared in an electrophoresis gel as described in (Dojnov and Vujčić, 2012) after protein separation by IEF, which was performed as described in Section 2.8.1.

2.9. Statistics

All reported experimental results (solid-state fermentation, enzyme assays and protein concentration) are calculated average measurements of repeated experimental runs (duplicates) with ±SD (standard deviation) less than 5% of the reported average. Six replications were performed per each isolate for statistical analysis of morpho-physiological studies. One-way analysis of variance among obtained data was tested by ANOVA, including comparison of means by *F*-test. The level of significance was at least $P < 0.05$ (Sokal and Rohlf, 1995). Statistical data analysis was performed using Statistica software for Windows 6.0 (Stat Soft Italia 1997).

3. Results

3.1. Morpho-physiological characteristics of isolates

All tested colonies contained snow white, lush, aerial mycelium with radial growth and without the ability to produce pigment colouring on PDA. The phialides of all tested isolates were flask-shaped, arising in veriticills of 2–5 on terminal branches. Green subglobose, smooth-walled conidia were produced on phialides after 2–5 days of incubation on PDA at 20 °C. Two distinct groups of *Trichoderma* isolates were found with regard to growth rate at 25 °C after 72 h (Table 2). In the first group growth rate was between 0.57 and 0.66 mmh^{-1} (TK1, TA1, and T42), and in the second, from 0.75 to 0.87 mmh^{-1} (TH2 and T55). Isolate TK1 was identified as *Trichoderma koningii* Oudem, characterized by mycelia with sweet coconut odour. Isolates TA1 and T42, with strong coconut-like odours were identified as *Trichoderma atroviride* P. Karsten. In the second group, isolates TH2 and T55, showed the same features as isolates from *Trichoderma* sp. group 3, as previously described by Kosanović et al. (2013) in a survey of green mould on mushrooms in Serbia, and they were classified in the same group. They were not identified to species level, as their characteristics were different from all known *Trichoderma* species. Isolate T10 was identified previously as *T. harzianum* Rifai (Kosanović et al., 2013).

All examined isolates are members of two clades: (a) clade *Rufa* – *T. koningii* (TK1) and *T. atroviride* (TA1 and T42) and (b) clade *Lixii/catoptron* – *T. harzianum* (T10) and *Trichoderma* spp. group 3 (TH2 and T55). The first clade is classified in the section *Trichoderma,* and the second in the section *Pachybasium B*. Both *T. atroviride* and *T. koningii* have demonstrated capacity for effective biological control due to their production of endochitinase enzymes and antifungal antibiotics, including aromatic pyrone antibiotics and peptides (Dodd et al., 2003). The majority of the analysed biocontrol *Trichoderma* species belong to *T.* section *Pachibasium*, suggesting that most of the potential biocontrol strains are associated with *T. harzianum* and its capacity to produce cell wall-degrading enzymes (Hermosa et al., 2004).

3.2. Phylogenetic relationship based on molecular criteria

Phylogenetic analysis of ITS1/ITS4 amplicon sequences showed that the analysed strains are distributed in three lineages which are in accordance with the differentiation obtained by studies of morpho-physiological features (Fig. 1). Four isolates out of the six studied were assigned to three different species T55 and TH2 clustered separately from other strains that were identified to species level and with strains designated by Kosanović et al. (2013) as *Trichoderma* sp. group 3, which is in agreement with data obtained by classical mycological studies. The results of both morpho-physiological and molecular studies confirmed that the isolates designated as *Trichoderma* sp. group 3 are different from the other tested isolates of the genus *Trichoderma*.

3.3. Analysis of SMC during fungal SSF

Approximately 33.2–36.4 g of substrate remained in each Erlenmayer flask after SSF. Cultivation of *Trichoderma* strains reduced the amount of SMC by nearly 30%. Changes occurring in the substrate–SMC were monitored by microscopic analysis of wheat straw as main substrate, and by determination of total protein and reducing sugars produced by SSF.

3.3.1. Microscopic analysis of wheat straw

A micrograph of raw wheat straw shows a regular pattern of parallel tubules with nodes and internodes between them. After

Table 2
Colony diameter (mm) of *Trichoderma* spp. isolates on potato dextrose agar depending on the temperature, three days after inoculation.

Species	Colony diameter (mm) Mean ± SE[a]						*F*	SEDs[b]
	Code	15 °C	20 °C	25 °C	30 °C	35 °C		
Trichoderma koningii	TK1	19.31 ± 4.30 c	30.75 ± 3.61 b	47.36 ± 3.11 a	48.11 ± 3.33 a	2.19 ± 2.19 d	1531.8	3.1
Trichoderma sp. group 3	TH2	23.20 ± 2.32 c	20.11 ± 1.91 c	62.55 ± 1.79 a	63.17 ± 8.5 a	32.22 ± 3.6 b	1729.5	4.03
Trichoderma atroviride	TA1	21.56 ± 4.9 c	19.25 ± 3.69 c	41.33 ± 4.88 a	35.20 ± 2.15 b	0.81 ± 2.26 d	741.6	3.45
Trichoderma atroviride	T42	27.11 ± 2.6 c	21.22 ± 3.36 d	44.83 ± 2.74 a	38.03 ± 3.33 b	1.17 ± 3.68 e	3229.5	2.9
Trichoderma sp. group 3	T55	15.86 ± 1.17 e	26.92 ± 1.2 c	53.67 ± 1.26 b	61.64 ± 1.2 a	18.61 ± 1.3 d	1601.0	1.12

[a] Data are means of six replicates ± SE, standard error of means.
[b] SEDs, standard error of differences; df, degrees of freedom = 25; $P < 0.001$.

growing of *A. bisporus* on wheat straw on mushroom farms interconnection of tubules disappeared—the strong adhesion connection between the tubules was disrupted (Fig. 2 centre) resulting in a porous structure.

The structure of wheat straw becomes even more disordered after cultivation of *Trichoderma* spp. and *A. niger* on SMC, as shown in Fig. 2. Major changes in wheat straw structure were visualized after growth of *T. harzianum* T10 and *Trichoderma* sp. group 3 TH2 isolates and minor changes were observed with isolate *Trichoderma* sp. group 3 T55. The tubules of wheat straw lost their parallel structure after growth of *Trichoderma* sp. group 3 TH2.

3.3.2. Parameters of fermentation efficiency

Polysaccharides in SMC (cellulose and hemicellulose) were hydrolysed by enzymes produced during growth of isolates of *Trichoderma* spp. and *A. niger*, which was followed by determination of the concentration of released reducing sugars. There were differences in the release of sugars between the examined fungal species (Fig. 3A). *T. konigii* TK1 released the highest level of reducing carbohydrates during SSF on SMC.

Levels of protein in crude extracts at the end of SSF are given in Fig. 3B. All *Trichoderma* spp. isolates produced higher levels of protein than *A. niger*. Isolates of *T. atroviride* T42 and *T. konigii* TK1 produced the highest levels of protein rendering them as potentially good enzyme producers.

3.4. Production of enzymes

3.4.1. Endocellulase production

All examined *Trichoderma* spp. isolates were better endocellulase producers than *A. niger* when SMC was used as substrate in SSF (Fig. 4). Levels of endocellulase produced by *Trichoderma* spp. that

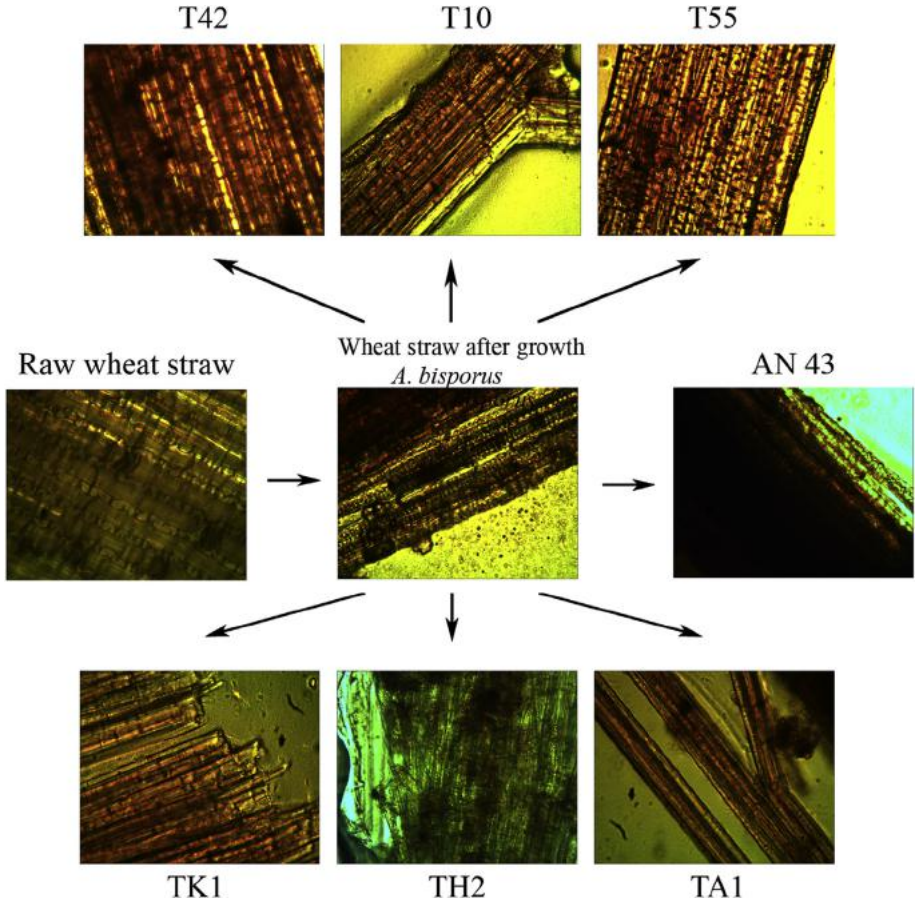

Fig. 2. Microscopic analyses of raw wheat straw and wheat straw after growth of *A. bisorus* and *Trichoderma* sp. and *A. niger*. TK1, TH2, TA1, T42, T10 and T55 are designations for *Trichoderma* strains; AN43 is the designation for *A. niger*, details are explained in Table 1.

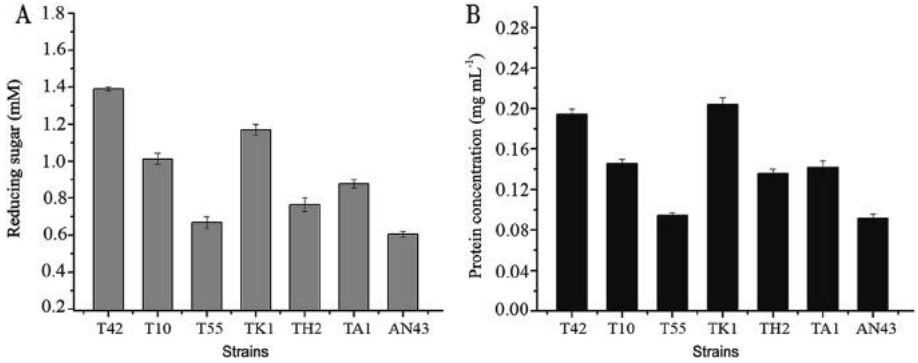

Fig. 3. Parameters of fermentation efficiency. A) Concentrations of proteins in crude SSF extracts (mg mL^{-1}); B) Concentration of reducing sugars in crude SSF extracts (mM). Each histogram column represents the mean of three independent assays ± SEM. TK1, TH2, TA1, T42, T10 and T55 are designations for *Trichoderma* strains; AN43 is the designation for *A. niger*, details are explained in Table 1.

was also isolated from mushroom farms, but cultivated with conventional methods, were lower than levels of endocellulase produced in our study, Fig. 4A, histogram T* (Chandel et al., 2013). *T. atroviride* T42, obtained from commercially grown *A. bisporus*, produced the highest level of endocellulase which is particularly evident from results measuring specific activity (Fig. 4A).

Zymographic detection of endocellulase isoforms is in accordance with enzymatic assays, Fig. 4A. Isolate *T. atroviride* T42 produced the highest number of endocellulase isoforms, as much as 11 within a wide range of pI values, of which isoforms 4, 5, and 9 showed the most intense bands (Fig. 4B). Both isoform 6 and 7 showed the most intense bands in the isolate *T. koningii* TK1.

3.4.2. β-glucosidase production

Three out of seven isolates examined here were able to produce β-glucosidase. The highest level of β-glucosidase was produced by isolate *T. koningii* TK1, as confirmed by detection of most intense isoform bands and presence of 5 isoforms on the zymogram (Fig. 5B).

3.4.3. Xylanase production

All fungal isolates examined here produced xylanase, moreover, all *Trichoderma* spp. isolates produced higher levels of xylanase than *A. niger*. Levels of xylanase produced by *Trichoderma* spp. isolates cultivated on residues of melon and pepper plant (Guisado et al., 2004) were lower than levels of xylanase produced by isolates investigated here (Fig. 6A, histogram T*).

Isolate T42 *T. atroviride*, a good cellulase producer, also produced high levels of xylanase with 12 isoforms of wide pI range (Fig. 6B). A similar number of isoforms with a wide pI range has been reported in fungal xylanase producers. Based on measurements of specific activity the isolate *T. harzianum* T10 produced the highest levels of xylanase.

3.4.4. Amylase production

The majority of *Trichoderma* spp. wild isolates (4 of 6) were better amylase producers than *A. niger* (Fig. 7). Production of amylase is not expected considering the results shown in Fig. 7A. Isolate *T. harzianum* T10 produced the highest level of amylase, in addition to the highest xylanase production level already observed. Amylase isoforms were detected in *T. koningii* TK1, *T. atroviride* TA1 and *T. atroviride* T42 isolates.

4. Discussion

Reduction of SMC by 30% through its use for cultivation of *Trichoderma* spp. presents a significant amount from an environmental point of view. This confirms our assumption that usage of SMC as substrate for fungal enzyme production is a promising method for reduction of its disposal and storage.

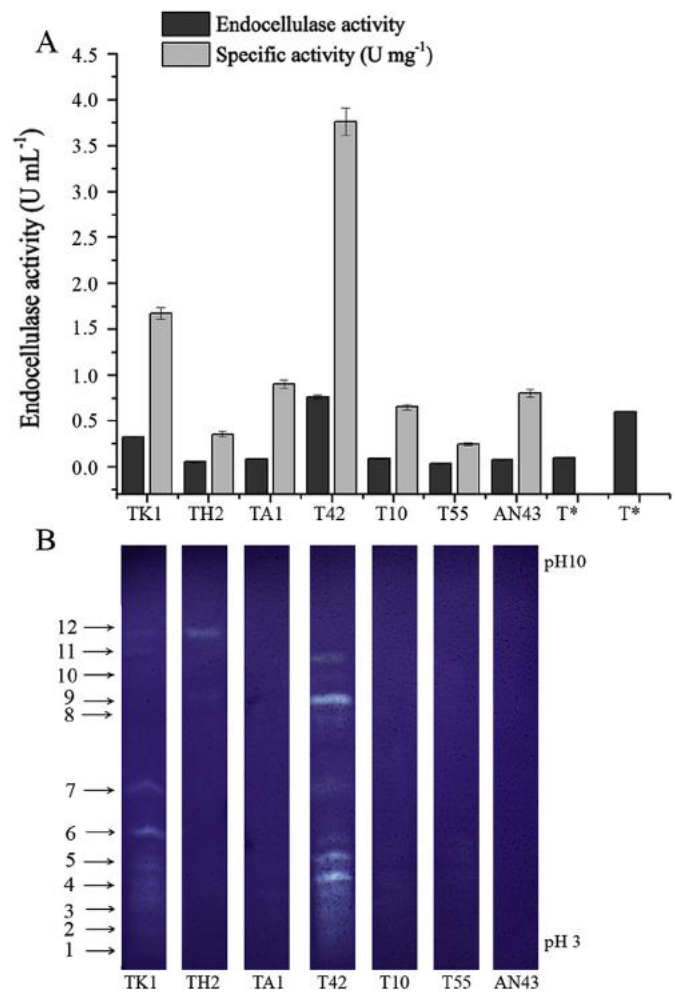

Fig. 4. Production of endocellulase in SSF using six wild type strains of *Trichoderma* sp. (TK1, TH2, TA1, T42, T10, T55) and *A. niger* (AN43) on SMC. A) Endocellulase activity (U mL^{-1}) and specific endocellulase activity (U mg^{-1} of protein) in crude SSF extracts. Each histogram column represents the mean of three independent assays ± SEM. B) Zymographic detection of endocellulase isoforms. TK1, TH2, TA1, T42, T10 and T55 are designations for *Trichoderma* strains; AN43 is the designation for *A. niger*, details are explained in Table 1. T* is the designation for *Trichoderma* sp. isolates adopted from Chandel et al. (2013). Arrows indicate the positions of isoforms.

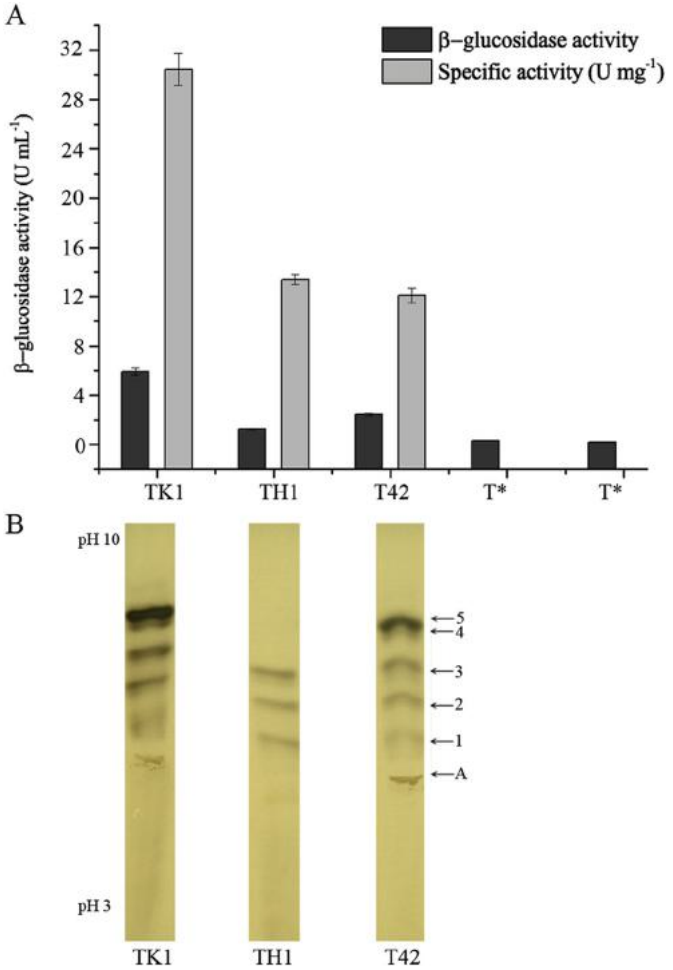

Fig. 5. Production of β-glucosidase in SSF using wild type strains of *Trichoderma* sp. (TK1, TH2 and T42) on SMC. A) β-glucosidase activity (U mL^{-1}) and specific β-glucosidase activity (U mg^{-1} of protein) in crude SSF extracts. Each histogram column represents the mean of three independent assays ± SEM. B) Zymographic detection of β-glucosidase isoforms. TK1, TA1 and T42 are designations for *Trichoderma* strains, details are explained in Table 1. T* is the designation for *T. viride* adapted from Adsul et al. (2004). Arrows indicate the positions of isoforms.

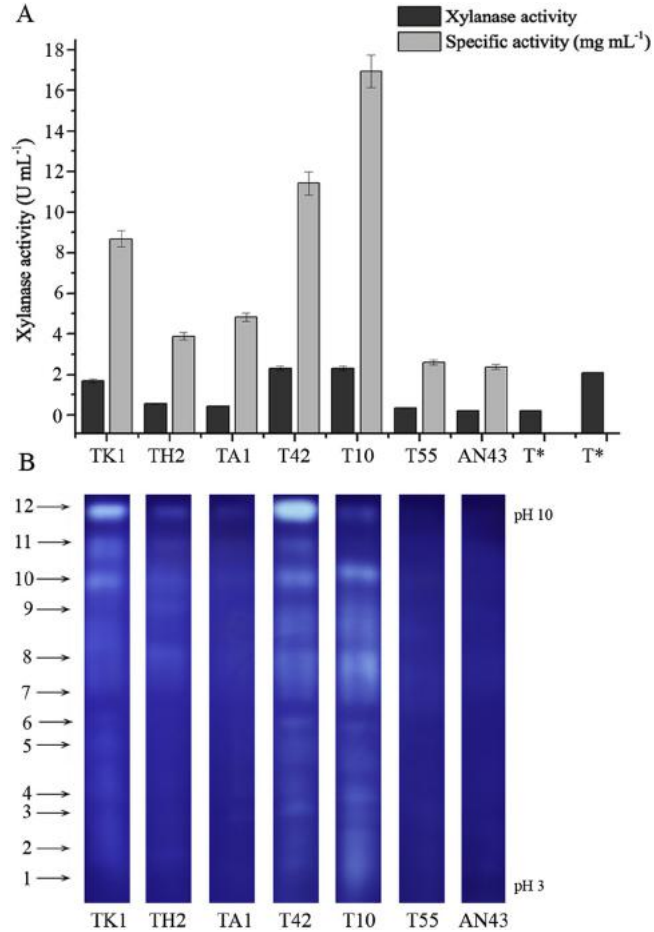

Fig. 6. Production of xylanase in SSF using six wild type strains of *Trichoderma* sp. (TK1, TH2, TA1, T42, T10, T55) and *A. niger* on SMC. A) Xylanase activity(U mL^{-1}) and specific xylanase activity (U mg^{-1} of protein) in crude SSF extracts. Each histogram column represents the mean of three independent assays ± SEM. B) Zymographic detection of xylanase isoforms. TK1, TH2, TA1, T42, T10 and T55 are designations for *Trichoderma* strains; AN43 is the designation for *A. niger*, details are explained in Table 1. T* is the designation for *Trichoderma* sp. isolates adapted from Guisado et al. (2004). Arrows indicate the positions of isoforms.

Observed changes in wheat straw structure indicate hydrolysis of carbohydrates or lignocellulotic material. It is known that the structure of wheat straw changes due to removal of lignin by *A. bisporus* (Zhang et al., 2002), visible on micrographs as disrupted connections between tubules. The partially hydrolized, porous wheat straw used in this study is a good substrate for *Trichoderma* spp. and *A. niger*. Therefore, mushroom breeding can be considered as pre-treatment of wheat straw for further *Trichoderma* or *Aspergillus* cultivation. Adhesion of *Trichoderma* and *A. niger* biomass on wheat straw leads to further changes in wheat straw structure. Changes of wheat straw tubules may be a consequence of hydrolysis of cellulose by enzymes from the cellulolytic complex (Hansen et al., 2011). Wheat straw changes can be visualized by the use of more detailed techniques, such as scanning electron and atomic force microscopy (Kristensen et al., 2008), as well. However, optical microscopy has proven to be a good qualitative indicator of substrate change. It can be useful as an affordable screening tool for the selection of fungal isolates, eliminating or reducing the need for expensive instruments such as a scanning electron microscope.

The concentration of reducing sugars at the end of the SSF process is not in direct correlation with the level of produced enzymes due to the fact that the fungi metabolized it. Filamentous fungi are capable of producing large amounts of protein, although the production level of any specific protein of interest in naturally occurring populations is usually too low for commercial exploitation (Punt et al., 2002). Quantification of total produced protein in fungal SSF may indicate the level of a specific enzyme and enable calculation of specific enzymatic activity which is used for comparison of production levels of different fungal isolates. This approach revealed that three different fungi (*T. atroviride, Trichoderma konigii and T. harzianum*), identified on species level, produced different amounts of protein.

Endocellulase (EC 3.2.1.4) is the key enzyme in the cellulase complex and hydrolysis of 1,4-β-D-glycosidic linkages in cellulose, lichenin and cereal beta-D-glucans. *Trichoderma* and *Aspergillus* are the most recognized fungi producers of the cellulase complex (Bansal et al., 2012). As the best producer of endocellualse, isolate,*T. atroviride* T42 can be regarded as a more promising cellulase producer than *Trichoderma reesei* RUT C30, the most frequently utilized fungal cellulase producer (Kovács et al., 2008). The highest level of specific endocellulase activity in *T. atroviride* T42 indicates that this enzyme is most abundant in the produced protein pool.

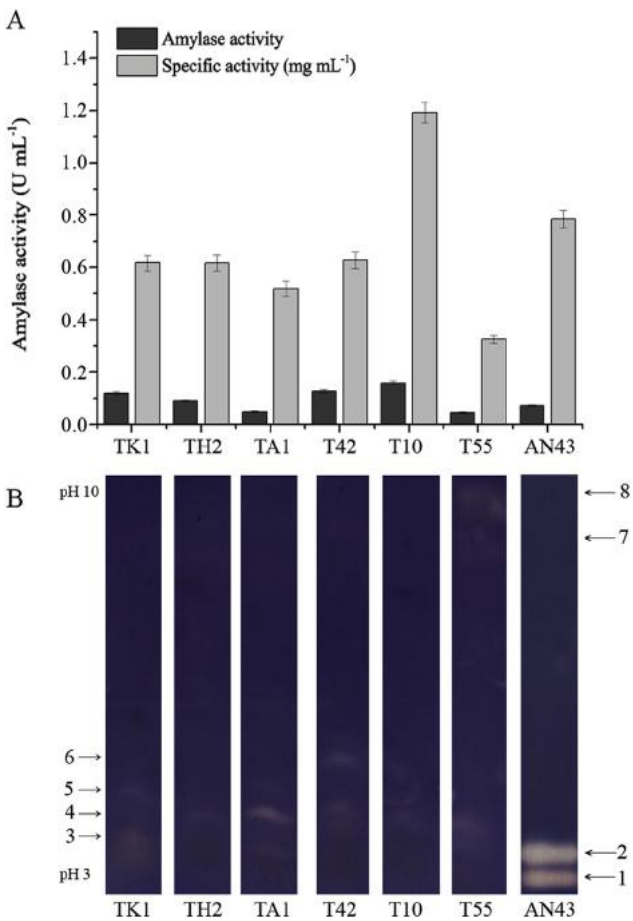

Fig. 7. Production of amylase in SSF using wild type strains of *Trichoderma* sp. (TK1, TH2, TA1, T42, T10 and T55) and *A. niger* on SMC. A) Amylase activity (U mL^{-1}) and specific amylase activity (U mg^{-1} of protein) in crude SSF extracts. Each histogram column represents the mean of three independent assays ± SEM. B) Zymographic detection of amylase isoforms. TK1, TH2, TA1, T42, T10 and T55 are designations for *Trichoderma* strains; AN43 is the designation for *A. niger*, details are explained in Table 1. Arrows indicate the positions of isoforms.

It is reported that some fungi can produce various endocellulase isoforms depending on the presence of metabolites (Nazir et al., 2010), or protease activity in fermentation extracts (Badhan et al., 2007). Isolate *T. atroviride* T42 produced as many as 11 isoforms, indicating a very high ability to adapt to a wide range of substrates and fermentation conditions. Cellulase isoform production in fungi can also be a consequence of transcription of different genes or post-transcriptional modification of enzymes (Willick and Seligy, 1985), which may explain the presence of different major isoforms in isolate *T. koningii* TK1.

β-glucosidase or cellobiase (EC 3.2.1.21) hydrolyses cellobiose, the exocellulase product, into free monosaccharides of glucose. It is the end point enzyme in complex hydrolysis of cellulose and it is a key component in bioethanol production from biomass via enzymatic route. Activity of β-glucosidase is limited by product inhibition (Singhania et al., 2010), this may be the reason for its lack in many cellulase production processes, including industrial hypercellulolityc mutant *T. reesei* RUT C30 (1% of total protein). Isolate T42 *T. atroviride* also produced β-glucosidase as part of the cellulase complex, which makes it even more significant as producer. Three isolates of *Trichoderma* spp. were able to produce β-glucosidase with different isoform profiles and a higher level of β-glucosidase than produced by *Trichoderma viride* on pre-treated sugar cane bagasse (Adsul et al., 2004). The presence of β-glucosidase isoforms in fungi depending on different SSF substrates has already been reported (Singhania et al., 2011).

Xylanase (EC 3.2.1.8) hydrolyses β-1,4-xylan,the most abundant component of hemicellulose, into xylose. Hemicellulose appears to hinder the approach of cellulases to cellulose fibres further explaining the importance of xylanase. Simultaneous production of cellulase and xylanase obtained by the examined strains is highly significant because it allows for total hydrolysis of lignocellulose biomass (Polizeli et al., 2005; Berlin et al., 2006; Faraco, 2013).

Amylase (EC 3.2.1.) hydrolyses the α-1.4-glycosidic bond in starch. *Aspegillus* is the most important fungal producer of amylase enzymes (Pandey et al., 1999). However, *Trichoderma* spp. produced higher amounts of amylase than *A. niger*. The presence of three amylase isoforms in isolates provides for differential substrate specificity.

Screening of fungi strains is the first step in the testing of serviceability of a material as potential substrate for production of industrially important enzymes. Our assumption was that *Trichoderma* spp. isolates obtained from mushroom-growing facilities could present good producers of cellulose- and hemicellulose-degrading enzymes, since compost may well be their natural environment. In support of this assumption, Williams et al. (2003) showed that antagonism of *T. harzianum* to *A. bisporus* is not primarily the result of mycoparasitism. The key factor seems to be competitive saprophytic ability of *T. harzianum* based on its capability to produce compost-degrading enzymes, cellulase and xylanase (Williams et al., 2003). The results of this study confirmed that assumption as well.

One of the examined isolates, *T. atroviride* T42 isolated from *A. bisporus* substrate, can be defined as a promising producer strain. Cellulase and xylanase are usually coproduced in *Trichoderma* spp. because their induction is closely related and based on induction of one enzyme which hydrolyses the substrate into products that are inducers of another enzyme in the complex (Royer and Nakas, 1990). Isolate *T. atroviride* T42 is the most promising fungi strain because it produced endocellulase, β-glucosidase and xylanase at the highest levels in comparison with other examined isolates. In addition, it produced all examined enzymes in a high number of isoforms, facilitating successful adaptation to new substrates or any type of industrially specific fermentation conditions.

5. Conclusions

Results shown in this paper open a new course in future perspectives to overcome problems with disposal of SMC. The total amount of SMC was reduced by 30% using simultaneously produced, industrially important cellulolytic enzymes. Material that is lagging behind is a more suitable fertilizer for horticulture and can be reused. *Trichoderma* spp. isolates were better producers of all industrially important enzymes examined in this study than reference strain *A. niger* ATCC. One isolate, *T. atroviride* T42, can be pointed out as a promising producer after further improvements.

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